INCH-POUND
MIL-I-14729A (AR)
AMENDMENT 2
27 August 1990
SUPERSEDING
AMENDMENT 1
8 August 1973

MILITARY SPECIFICATION

INCREMENT, PROPELLANT, M90A1

This Amendment forms a part of Military Specification MIL-I-14729A (MU), dated 5 June 1969, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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Title, Delete "M90E1" wherever it appears and substitute "M90A1."

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3.3 "Strength of heat seal." Delete this paragraph in its entirety.

3.4 "Gross weight of propellant increment." Delete this paragraph in its entirety and substitute the following:

"3.4 Gross weight of propellant increment. The weight of the loaded projectile increment within each lot shall vary not more than plus three or minus three grains from the established increment weight (assessed propellant weight plus propellant bag tare weight) (see para. 4.2.3.2)."

Add the following paragraph:

"3.6 Weight of propellant. The weight of the propellant, loaded into the propellant bag, shall vary not more than plus three or minus three grains from the assessed propellant weight."

Add the following paragraph:

"3.7 Vacuum Test. The loaded, sewn and sealed propellant increments shall withstand a surrounding absolute pressure of 2 to 7 inches of mercury (i.e., 23 to 28 inches of mercury gage vacuum) for 5 seconds minimum without deflating. Increment shall be tested as specified in para. 4.3.2."

AMSC N/A FSC 1375

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4.2.2.1 Delete Critical Defect "1. Bag or bag seam... propellant can escape" and substitute "1. Bag or bag seam damaged or stitches missing to extent that propellant can escape."

Delete Major Defect "101. Bag or bag seam damaged...incomplete along any seam...Visual 01002."

Delete Major Defect "102. Stitches broken (on ends of bag that are not heat sealed)...Visual 01010."

Delete Minor Defect "205. Stitches broken or missing" and substitute "205. Stitches broken or unraveled along heat seal (broken stitch permitted at intersection of stitch and heat seal)."

Add Minor Defect "208. End folds not fully sewn (by transverse stitching)...01010."

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4.2.3 Delete paragraphs 4.2.3.1 and 4.2.3.2 in their entirety and add the following paragraphs:

"4.2.3.1 Weight of propellant. (see 3.6) Critical Defect (under minimum weight); Major Defect (over maximum weight)-Code No. 05001.-Prior to loading, the propellant shall be weighed 100 percent.

4.2.3.2 Gross weight of propellant increment. (see 3.4)-Critical Defect (under minimum weight); Major Defect (over maximum weight)-Code No. 04001.-The loaded propellant increments shall be weighed 100 percent. Any propellant increment which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with paragraph 4.2.4."

4.2.3.3, "Strength of heat seal.", delete this paragraph in its entirety.

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Add the following paragraph:

"4.2.3.5 Vacuum Test. Critical Defect-Code No. 09001-The propellant increments shall be tested 100 percent. If any increment fails to meet the requirement (see para. 3.7), it shall be rejected and removed from the lot. Testing shall be performed as specified in para. 4.3.2 using equipment in accordance with paragraph 4.2.4."

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4.3.1.1 "Strength of heat seal." Delete this paragraph in its entirety.

Add the following paragraph:

"4.3.2 Vacuum Test. Propellant increments shall be confined to slots 1 1/2 inches deep x 1/2 inches high x a minimum of 6 inches wide for B charges and a minimum of 9 inches wide for A charges. Units of slots may be integrated to comprise an increment rack. Increments, contained in the slots, shall be placed horizontally into a vacuum chamber. Increments shall be subjected to a surrounding absolute pressure of 2 to 7 inches of mercurly (i.e., 28 to 23 inches of mercury vacuum). Vacuum shall be drawn gradually and uniformly with respect to time. Vacuum shall be attained in no less than 10 seconds. The increment rack shall be rotated 90 degrees (under vacuum) to enable those defective (non-inflated) increments to fall free of their slots. The rack shall be held in this position for a minimum of 5 seconds. The increment rack shall then be rotated back to its original position. The rack will then contain acceptable increments only. Defective bags (leakers) will have dropped to the bottom of the vacuum chamber.

The rack shall be subjected to an initial and a continuous, Picatinny Arsenal approved, qualification procedure. This procedure shall consist of dumping the loaded rack a number of times without vacuum, and in accordance with the same dumping cycling time used for actual testing. Failure to release increments shall be reason for rack disqualification."

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* 5.1.1, Delete paragraph in its entirety and substitute:

"5.1.1 Level A. Propellant increments shall be packed in heat sealed barrier bags complying with Type I, Class E, of MIL-B-117. The barrier bag size shall be 22 1/2 inches long by 22 1/2 inches wide, nominal. Approximately 15 pounds of propellant shall be packed in each bag and closed by means of heat sealing. Two sealed bags shall be packed in a wooden box (Suggested inside box dimensions 14 1/2 inches by 7 1/4 inches.) conforming to Grade A, Type I, Class 1 or 3 of MIL-B-2427. Both ends, sides, top, and bottom of the wooden box shall be lined with 1/16 inch thick solid water resistance sheet material conforming to MIL-F-50499. The box shall be closed and strapped in accordance with MIL-B-2427. The net weight shall not exceed 30 pounds of propellant."

* 5.1.1.1, Add new paragraph as follows:

"5.1.1.1 Alternative Level A. Propellant increments shall be packed in propellant packing boxes M17, M18, M24 or M25 complying with dwg. 75-4-56, 76-4-59, 76-4-46, and 7549033 respectively. In lieu of air pressure test, two 4-MIL conductive polyolefin or polyethylene bags can be used for all containers listed herein. Each bag must be individually tied and shall be a lay flat syle bag made of conductive material in accordance with 5.1.2.1."

* 5.1.2, Delete paragraph in its entirety and substitute:

"5.1.2 Level B. Propellant increments shall be packed in a fiberboard carton conforming to requirements referenced on dwg. 7548645. The liner shall be a lay flat style bag made of conductive material in accordance with 5.1.2.1."

* 5.1.2.1, Add new paragraph as follows:

"5.1.2.1 Liners. Liners shall be made of conductive polyethylene or polyolefin sheet with a thickness of .004 inch minimum, surface resistivity shall not exceed 1000 ohms per centimeter, when tested in accordance with Methods 4041 and 5052 of FED-STD-406. Electrostatic bleed-off shall not

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exceed 2.00 seconds when tested in accordance with Method 4046 of FED-STD-101. The liner shall be closed by rolling down the top and taped (with pressure tape made in accordance with PPP-T-97 or MIL-T-43036) to prevent unrolling or it may be heat sealed. Details not specified herein shall comply with PPP-B-26."

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* 5.1.3, Delete paragraph in its entirety.

* 5.1.3.1, Delete paragraph in its entirety.

* 5.2.2, Delete paragraph in its entirety and substitute:

"5.2.2 Alternative Level A. The M17, M18 and M24 Containers shall be marked in accordance with dwg. 8858848. The M25 Container shall be marked in accordance with dwg. 8858577."

* 5.2.3, Delete paragraph in its entirety and substitute:

"5.2.3 Level B. The fiberboard carton or other container shall be marked in accordance with dwg. 9241693."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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